CS2201 Computer Organization & Architecture

L1: Introduction and Course Overview



John Jose

Associate Professor

Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

Few Important Information

- Instructor: John Jose
  - Office Room: H-201, Second Floor, CSE department
  - Webpage: http://www.iitg.ac.in/johnjose/ Email:johnjose@iitg.ac.in
- Teaching Assistants
  - Mr. Nikhil Singh Khatuwal * (n.khatuwal@iitg.ac.in)
  - Dr. Nidhika Chauhan (nidhi29.chauhan@gmail.com)
  - Ms. Rajeswari Suance P.S. (s.rajeshwari@iitg.ac.in)
  - Mr. Thirtha Pattanayak (p.tirtha@iitg.ac.in)
  - Mr. Mallika Chouhan (mallikachouh@iitg.ac.in)
- Lecture Slots and Venue and Course Management
  - C slot @ 5101 : Mon (10 AM), Tue (11 AM), Fri (9 AM)
  - MS Teams for course information, slides and updates

Grading

- Quiz-1: 21.08.2026 (Fri) [15%]
- Quiz-2: 16.10.2026 (Fri) [15%]
- Group Assignment:* 07.11.2026 (Sat) [10%]
- Mid Semester Exam: 16.09.2026 (Wed) [20%]
- End Semester Exam: 17.11.2026 (Tue) [30%]
- Attendance and Class Participation [6+4] [10%]

- Minimum mandatory attendance to avoid deregistration is 60%

Reference Books

- **Computer Organization and Design: The Hardware/Software Interface,** John L. Hennessy, David A. Patterson, Morgan Kaufman
- **Computer Architecture: A Quantitative Approach**, 6th Edition, John L. Hennessy, David A. Patterson, Morgan Kaufman
- **Computer Organization and Architecture,** William Stallings, Pearson Education India.
- **Advanced Computer Architecture,** Smruti Ranjan Sarangi, McGrawHill.
- **Computer Organization and EmbeddedSystems**, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, and Naraig Manjikian, McGrawHill.

General Policies

- 100% attendance is preferred.
- Once you miss the class you will lose the connectivity between topics.
- Be on time in attending lecture class.
- Introductory 5 minutes is very important for the day's discussion.
- Academic dishonesty cannot be tolerated.
- Everybody cannot score AA/AS. Do your best, Be sincere, Be open.
- It is not the marks but the effort that matters.
- I promise that you will enjoy this course.

Syllabus: Part A

- **Introduction to arithmetic circuits:** Review of Fixed-point representation, Booth's multiplication, restoring and non-restoring division circuits, character encoding basics, IEEE 754floating-point arithmetic circuits. **[4 hours]**
- **Instruction Set Architecture:** Principles of ISA design, instruction formats and encoding, addressing modes, Introduction to assembly language programming using a modern RISC ISA **[4 hours]**
- **Processor Design:** Single-cycle data path design and control unit, Introduction to 5-stage RISC instruction pipeline, pipeline hazards, hazard mitigation techniques, branch prediction. Introduction to static and dynamic scheduling, out-of-order execution and speculative execution **[10 lectures]**

Syllabus: Part B

- **Memory System:** Cache memory - mapping, cache replacement policies and write strategies, cache performance analysis, cache optimization techniques, Introduction to DRAM organization, Introduction to Non-Volatile Memory organization. High Bandwidth Memory **[10 hours]**
- **Input-Output Systems:** Introduction to hard disk organization, SSDs. Introduction to I/O transfer mechanisms, DMA, interrupt handling, I/O interfaces and buses, PCIe, PCIx, USB and CXL standards; **[4 hours]**
- **Introduction to Multicore architectures:** Superscalar processor and Multi-core designs, Introduction to GPU architecture and heterogeneous computing. Introduction to domain specific accelerator. **[4 hours]**

Applications and Systems

Applications run on a computer system that has processor(s) connected to high-speed memory.











Course Objective

- **Understand** working principles of basic components that constitute a computer – processor, memory, storage, I/O and interconnect.
- **Learn** computer architecture with an emphasis on system design, performance and analysis.
- **Understand** and **analyze** events happening at hardware level.
- **Elevate** thinking process to the level of performance improvement techniques in computer architectures.
- **Explore** future directions in computer architecture research.

Relevance of the Course

- ML applications will dominate future computing systems.
- Effective leverage of AI/ML can be possible only with ML friendly hardware.
- AI Engines and ML processors will rule the future world.
- We need good computer architects and processor design engineers with exposure to hardware design to cater the growing demand of skilled personnel in this domain.
- India needs to scale up manpower with passion and skill to contribute to building compute systems that augment AI growth.

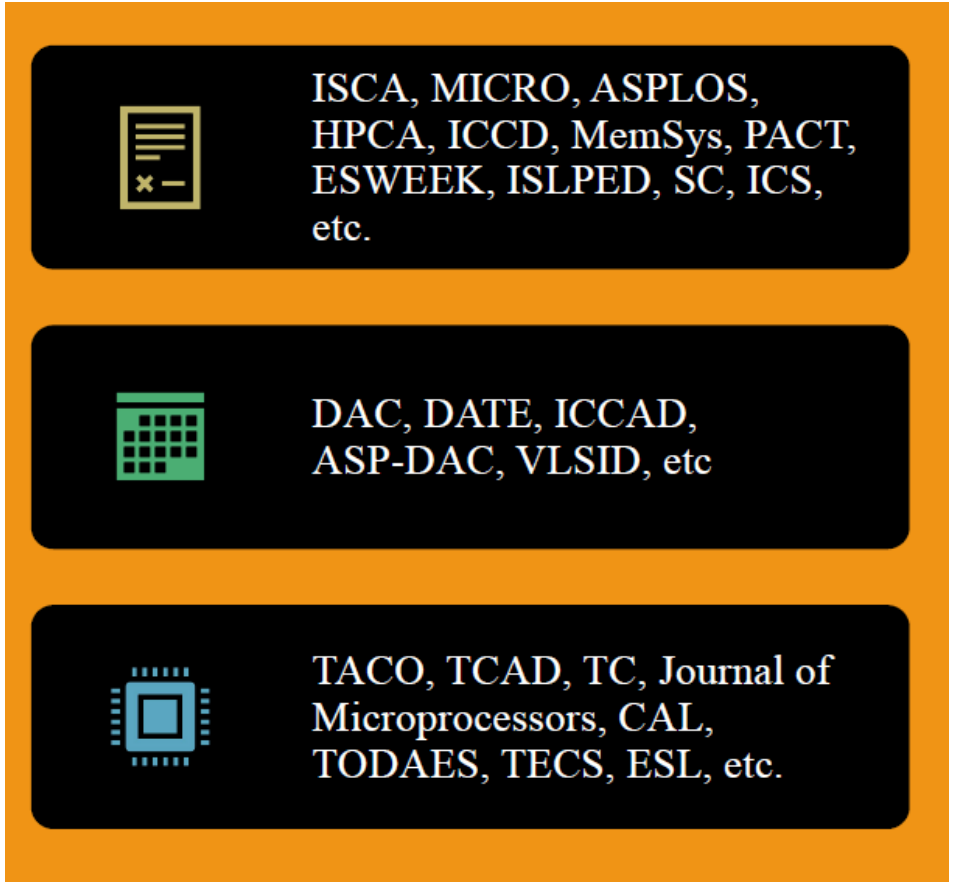
Career Paths in Computer Architecture

- **Performance Jobs:** Working on simulators and projecting system performance using benchmarks. They also include running workload on real hardware and evaluating performance.
- **Hardware Design Jobs**: Working on Verilog/VHDL/System Verilog to build real hardware logic.
- **Verification Jobs:** Functional and performance verification roles.
- **Analyst Jobs:** Do comparative analysis of various system parameters.
- **Architect Jobs:** Top level system design by integrating various microarchitectural components

Companies in Computer Architecture

Year of Start	Company	Year of Release First Processor
1899	Nippon Electrical Company (NEC)	1984
1911	International Business Machines (IBM)	1960s
1927	Galvin Manufacturing Corporation (Motorola in 1947)	1974
1951	Texas Instruments (TI)	1971
1968	Integrated Electronics (Intel)	1971
1969	Advanced Micro Devices (AMD)	1971
1993	Nvidia	GeForce 256 in 1999
1985	Array Technology Inc (ATI) now AMD Radeon	1987
1990	Advanced RISC Machines	Qualcomm, Apple, NXP, TI, Hitachi, Samsung, Marvell, Mediatek, Nvidia, and many more use ARM IP to build own chips.

Where to see modern computer architecture?



Relationship with COA & Other Subjects

Programming & Data Structures

- Smart programmers understand systems well
- Designing new language
- Smart Architects are Expert Programmers
- Ability to view components as objects
- Parallel programming
- Debugging skills

Algorithm Design

- CA is the study of Hardware Algorithms!
- Time Complexity (Performance Wall)
- Space Complexity (Area Wall)
- Design Complexity
- Power Complexity (Power Wall)
- Security Complexity, etc.

Relationship with COA & Other Subjects

Digital Logic & Design

- Fundamentals to designing systems
- Understanding of Gates, Combinational Circuits, and Sequential Circuits
- Physics of the devices will help building expertise

Mathematics

- Elementary Mathematics
- Theory of Computation: Mealy and Moore machines are basis of designing complex systems
- Queuing Theory: For experts

Software Engineering

- Hardware Engineering is done by writing code!
- Analysis, design, verification and testing as per SDLC

Relationship with COA & Other Subjects

Operating Systems

- System software that manages the hardware
- Designing and understanding OS requires understanding of architecture
- Application optimization at hardware software boundary

Computer Networks

- Fundamentals of Input Output concepts
- Fundamentals of network devices
- Data format device organization
- Packet processing and network delay

Compilers

- Back-end of the compiler deals with code generation and code optimization
- Code generation is target hardware specific

Relationship with COA & Other Subjects

Artificial Intelligence

- AI wave is supported by specific hardware
- We use predictors - Branch predictors & prefetchers that are needed heavily on AI tasks
- AI efficiency is done through hardware accelerators

Data Science

- Computer Hardware can produce more data than any other systems
- Data analytics and performance finetuning at hardware is getting big traction

Secure Systems Engineering

- Hardware security
- Secured System Management
- Access Control

Who should take this course?

This course is for those who are willing to take up the risk **(enjoy the beauty)** in understanding how processor, memory and communication technologies of a high-end computer operate by exploring its architecture.

It is not the **destination** (final grade in the course), but the unique experience in the **journey** (discussions in lecture videos, tutorial sessions, deeper insights gained while solving assignments etc) that is important.

So enjoy the ride !!!

MARS Lab

- Members: https://www.iitg.ac.in/johnjose/students.html

- MARS Lab Video: **https://tinyurl.com/27gkx8xo**

Student Data Form

https://forms.gle/t8iH81BYr6T4Nuum7



[Mandatory to be submitted before 26.07.2026]



johnjose@iitg.ac.in

http://www.iitg.ac.in/johnjose/

LinkedIn: https://www.linkedin.com/in/john-jose-bb8431106/